

Solution Architecture

Date: 30 September 2026 **Team ID:** 6ab3e57da238dc999eb3d44d **Project Name:** EduGenie – AI-Powered Educational Assistant

Maximum Marks: 5 Marks

Architecture Flow

Presentation / Client Layer	Application Layer	AI / External Service	Storage / Configuration
HTML template (index.html) CSS (style.css) JavaScript (app.js)	FastAPI (main.py) Routes: /qa, /explain, /summarize, /quiz, /learn/recommendations Feature modules: qna.py, explanation_module.py, summary_module.py, quiz_module.py, learning_path.py	ai_service.py Google Gen AI SDK Gemini / configured model Demo fallback if API is unavailable	No persistent database in current release Environment variables (.env) for API configuration

Component Description

Component	Description / Role	Technology
Presentation Layer	Collects user input and presents results.	HTML/CSS/JavaScript/Jinja2
API Layer	Validates and routes requests; serves the frontend.	FastAPI + Pydantic
Feature Modules	Each workflow owns its prompt and feature logic.	Python modules
AI Service	Centralizes provider setup and generation behavior.	google-genai + Gemini
Quiz Validation	Cleans model output, parses JSON and checks list structure.	Python json/re
Configuration	Keeps API key and model name outside source code.	python-dotenv / environment variables